

Ex.no:4b	EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION

Aim:

To perform exploratory data analysis on the E-Commerce dataset by inspecting, filtering, sorting, grouping, performing statistical and correlation analysis, and visualizing the results using Pandas, Matplotlib, and Seaborn.

Algorithm:

1. Import the required Pandas, NumPy, Matplotlib, and Seaborn libraries.
2. Upload and read the E-Commerce dataset into a DataFrame.
3. Display the first few records, dataset shape, and data types.
4. Generate the statistical summary of the dataset.
5. Filter products based on unit price and customers based on country.
6. Sort the records based on unit price in descending order.
7. Group the data by country and calculate the average unit price.
8. Calculate the mean, median, mode, and standard deviation of unit price.
9. Calculate the correlation between selected numerical variables.
10. Display the correlation matrix using a heatmap.
11. Create a bar chart showing average unit price by country.
12. Create a scatter plot to analyze the relationship between quantity and unit price.
13. Display and interpret the obtained results.

Program:

```
# Import libraries
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

```
file_name = list(uploaded.keys())[0]
```

```
df = pd.read_csv(file_name)
```

```
print(df.head())
```

```
print("\nDataset Shape")
```

```
print(df.shape)
```

```
print("\nData Types")
```

```
print(df.dtypes)
```

```
print("\nStatistical Summary")
```

```
print(df.describe())
```

```
high_price = df[df["unit_price"] > 500]
```

```
print("\nHigh Unit Price Products")
```

```
print(high_price.head())
```

```
india = df[df["country"] == "India"]
```

```
print("\nCustomers from India")
```

```
print(india.head())
```

```
sorted_df = df.sort_values(
```

```
        by="unit_price",
        ascending=False
    )

    print("\nTop Products by Unit Price")
    print(sorted_df.head())

    print("\nMean:", df["unit_price"].mean())

    print("Median:", df["unit_price"].median())

    print("Mode:")
    print(df["unit_price"].mode())

    print(
        "Standard Deviation:",
        df["unit_price"].std()
    )

    country_avg = (
        df.groupby("country")["unit_price"]
        .mean()
        .sort_values(ascending=False)
        .head(10)
    )
```

```
print("\nAverage Unit Price by Country")

print(country_avg)

country_avg.sort_values(
    ascending=False
).plot(kind="bar")

plt.title("Top Countries by Average Unit Price")
plt.xlabel("Country")
plt.ylabel("Average Unit Price")

plt.show()

corr = df[
    [
        "age",
        "cancellations_count",
        "unit_price",
        "quantity",
        "purchase_frequency"
    ]
].corr()

print("\nCorrelation Matrix")
```

```
print(corr)
```

```
print(  
    "\nCorrelation between Quantity and Unit Price:",  
    df["quantity"].corr(df["unit_price"])  
)
```

```
plt.figure(figsize=(10, 6))
```

```
sns.heatmap(  
    corr,  
    annot=True,  
    cmap="coolwarm",  
    fmt=".2f"  
)
```

```
plt.title("Correlation Matrix of E-Commerce Factors")
```

```
plt.show()
```

```
sns.scatterplot(  
    data=df,  
    x="quantity",  
    y="unit_price"  
)
```

```
plt.title("Quantity vs Unit Price")
```

```
plt.xlabel("Quantity")
```

```
plt.ylabel("Unit Price")
```

```
plt.show()
```

```
plt.figure(figsize=(10, 5))
```

```
plt.plot(
```

```
    df.index[:20],
```

```
    df["unit_price"].head(20),
```

```
    marker="o"
```

```
)
```

```
plt.title("Unit Price Trend")
```

```
plt.xlabel("Product Record")
```

```
plt.ylabel("Unit Price")
```

```
plt.show()
```

```
plt.figure(figsize=(8, 5))
```

```
plt.hist(
```

```
    df["unit_price"],
```

```
    bins=10  
)
```

```
plt.title("Unit Price Distribution")  
plt.xlabel("Unit Price")  
plt.ylabel("Number of Products")
```

```
plt.show()
```

```
plt.figure(figsize=(8, 5))
```

```
plt.boxplot(df["unit_price"])
```

```
plt.title("Unit Price Box Plot")  
plt.ylabel("Unit Price")
```

```
plt.show()
```

Output:

High Unit Price Products

order_id	customer_id	age	product_id	country	signup_date	\
1	ORD0002	CUST1002	26	PROD002	Pakistan	06-10-2023
8	ORD0008	CUST1008	51	PROD008	Pakistan	6/25/2023
11	ORD0011	CUST1011	24	PROD011	Canada	02-12-2022
26	ORD0026	CUST1026	34	PROD026	USA	07-06-2023
28	ORD0028	CUST1028	42	PROD028	USA	5/25/2023

last_purchase_date	cancellations_count	subscription_status	order_date	\
1	09-04-2023	3	cancelled	05-12-2025
8	12/26/2024	4	paused	12/14/2023
11	03-12-2025	1	active	5/30/2025
26	06-02-2025	1	active	6/10/2024
28	01-09-2025	5	paused	6/13/2025

unit_price	quantity	purchase_frequency	preferred_category	product_name	\
1	665.14	2	44	Electronics	Hoodie
8	1525.00	1	32	Electronics	Nail Polish
11	940.17	1	24	Beauty	Headphones
26	1075.00	1	31	Sports	Football
28	1702.39	1	14	Clothing	Sneakers

category	gender
1	Clothing Female
8	Beauty Female
11	Electronics Female
26	Sports Female
28	Clothing Male

Customers from India

order_id	customer_id	age	product_id	country	signup_date	\
3	ORD5003	CUST1003	54	PROD203	India	7/30/2023
4	ORD5004	CUST1004	50	PROD204	India	12-09-2020
19	ORD5019	CUST1019	51	PROD219	India	9/14/2022
25	ORD5025	CUST1025	49	PROD225	India	11-07-2021
28	ORD5028	CUST1028	66	PROD228	India	06-02-2021

last_purchase_date	cancellations_count	subscription_status	order_date	\
3	2/20/2024	4	paused	9/19/2024
4	9/14/2024	0	active	08-08-2024
19	8/13/2025	0	paused	10/27/2024
25	06-09-2022	3	cancelled	9/26/2022
28	04-03-2025	1	active	1/17/2022

unit_price	quantity	purchase_frequency	preferred_category	product_name	\
3	84.66	4	1	Sports	Conditioner
4	62.66	7	35	Sports	Smartwatch
19	199.58	8	41	Clothing	Refrigerator
25	142.89	5	40	Clothing	Sneakers
28	149.65	8	45	Electronics	Sofa

category	gender
3	Beauty Male
4	Electronics Male
19	Home Female
25	Clothing Male
28	Home Male

Top Products by Unit Price

	order_id	customer_id	age	product_id	country	signup_date	\
1273	ORD6273	CUST2273	25	PROD1473	Pakistan	3/31/2021	
984	ORD5984	CUST1984	25	PROD1184	India	3/14/2021	
1874	ORD6874	CUST2874	50	PROD2074	India	5/21/2021	
42	ORD5042	CUST1042	57	PROD242	Germany	5/23/2023	
684	ORD5684	CUST1684	40	PROD884	India	4/13/2022	

	last_purchase_date	cancellations_count	subscription_status	order_date	\
1273	7/29/2021	5	active	4/21/2021	
984	06-10-2022	1	active	6/18/2025	
1874	2/16/2025	0	cancelled	07-05-2025	
42	3/15/2025	4	paused	8/25/2023	
684	03-04-2025	1	active	06-11-2025	

	unit_price	quantity	purchase_frequency	preferred_category	\
1273	1991.63	1	31	Beauty	
984	1982.82	1	7	Sports	
1874	1955.81	1	30	Electronics	
42	1954.64	1	36	Home	
684	1951.85	1	24	Home	

	product_name	category	gender
1273	Dress	Clothing	Female
984	Sneakers	Clothing	Male
1874	Skirt	Clothing	Male
42	Cricket Bat	Sports	Other
684	Conditioner	Beauty	Male

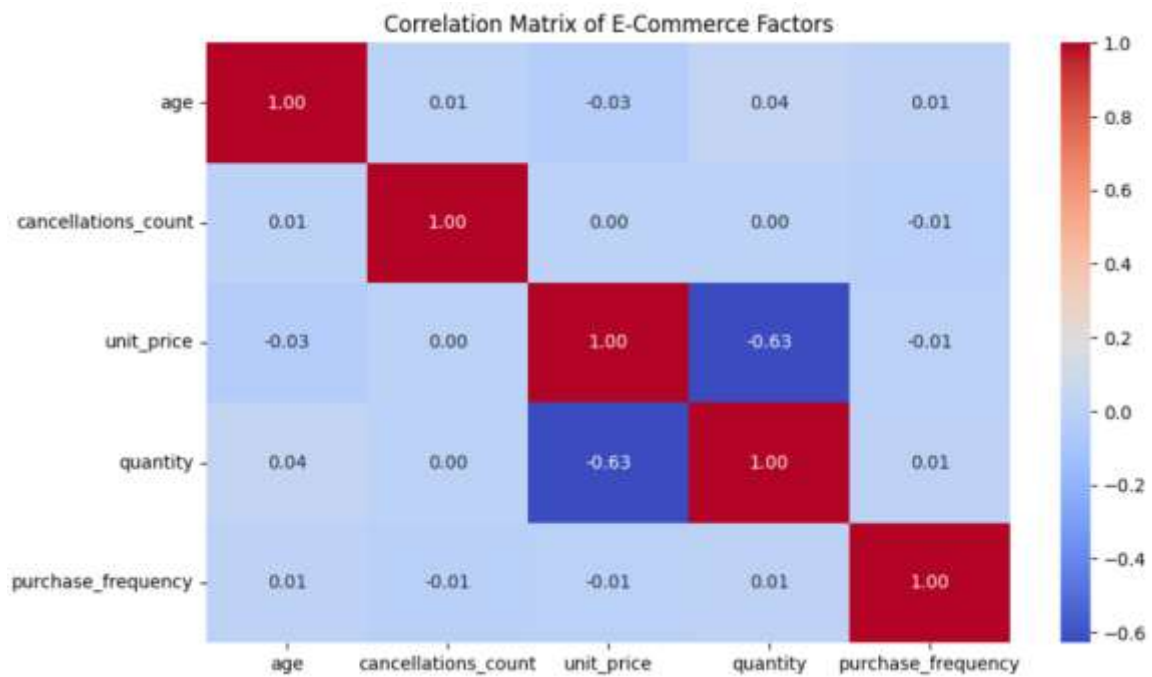
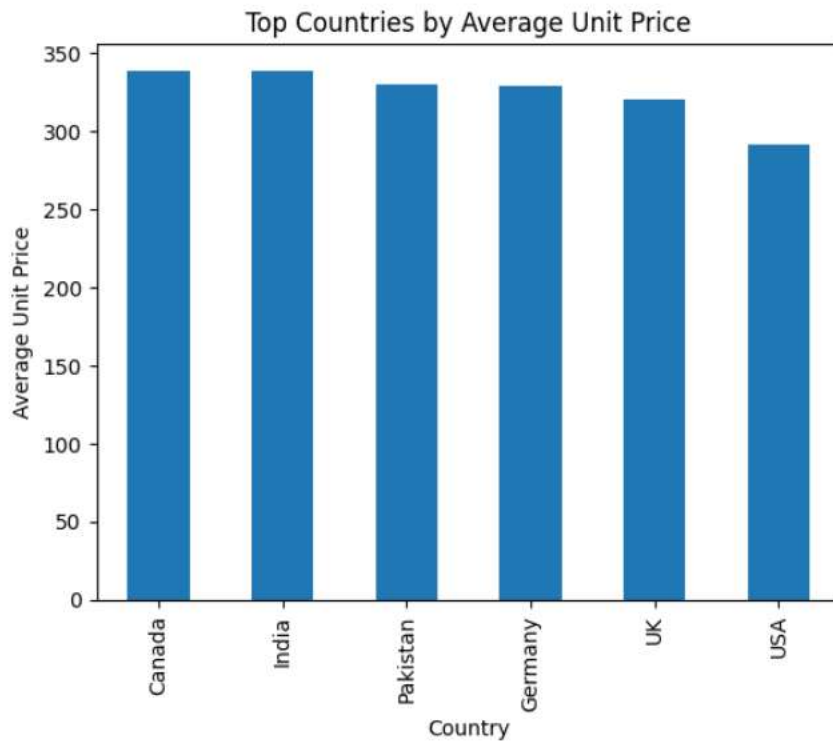
Mean: 324.71946

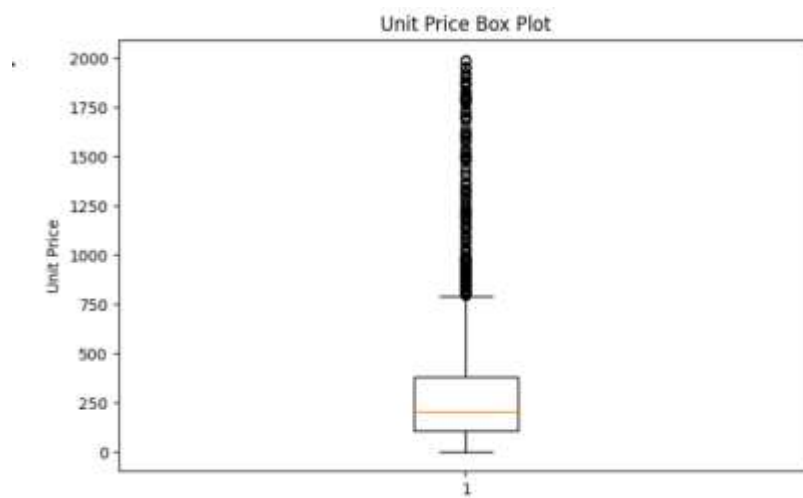
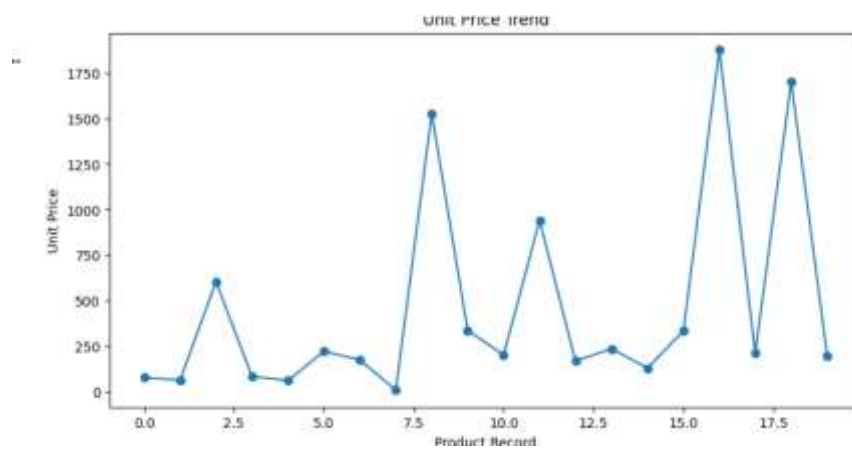
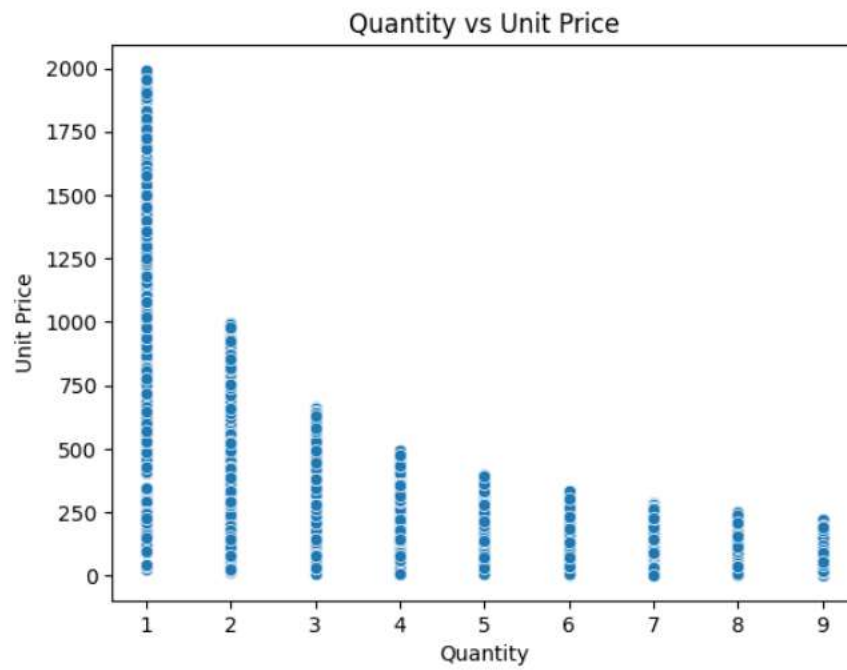
Median: 206.515

Mode:

0 43.76

Name: unit price, dtype: float64





Result: